Solutions:

(i) $\int x^7\,dx = \frac{x^{7+1}}{7+1} = \frac{x^8}{8} + c$

(ii) $\int \frac{1}{x^6}\,dx$

$= \int x^{-6}\,dx$

$= \frac{x^{-6+1}}{-6+1} = \frac{x^{-5}}{-5}$

$= -\frac{1}{5x^5} + c$

(iii) $\int \sqrt{x}\,dx$

$= \int x^{1/2}\,dx$

$= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1}$

$= \frac{x^{\frac{3}{2}}}{\frac{3}{2}}$

$= \frac{2}{3}x^{\frac{3}{2}+c}$

(iv) $\int (x^5 - 4x^2 + 3x + 2)\,dx$

$= \left(\frac{x^6}{6}\right) - 4\left(\frac{x^3}{3}\right) + 3\left(\frac{x^2}{2}\right) + 2x + c$

The above discussed integrals are known as improper integrals or indefinite integrals . For the proper or definite integrals we have the limiting points at both sides. These are called the lower limit and the upper limit of the integral.

This integral $\int f(x)\,dx$ is an indefinite integral. Integrating the same function with in the given limits a and b is known as the definite integral. We write this in symbol as

$\int_a^b f(x)\,dx$ =k(a constant value)

In a definite integral where a is known as the lower limit and b is known as the upper limit of the definite integral. To find the value of definite integral, we do as follows:

Suppose $\int f(x)\,dx = F(x)$

Then $\int_a^b f(x)\,dx = F(b) - F(a)$